

Gila Monster — Constant-Current Biphasic (or Monophasic) Stimulator v1.0

Firmware Bench Test Report

Unit under test: Gila Monster dual-port firmware, output A on D7 and output B on D8, Arduino Uno R3 (ATmega328P, 16 MHz).

Instruments: digital oscilloscope, two DC-coupled 1 M Ω probes (CH1 on D7, CH2 on D8), triggered on channel A; firmware session report over the serial port (Comm, 9600 baud, 8-N-1).

Session length: 300 s (Timer = 300,000 ms) in all tests.

Pulse pair definition: port A rises, Pulse, port A falls, Delay, port B rises, Pulse, port B falls. The two pulses never overlap and share the same programmed width.

Cycle constraint enforced by the menu: $2 \times PW + Delay + 100 \mu s \leq T$ (Continuous and Burst) or $\leq ITI_{min}$ (NPS).

1. Delay accuracy

#	Configuration	Regime	Measured
1A	20.00 Hz, PW 10.00 ms, Delay 25 μs	rising edge exact	25 μs , no dispersion; equal widths
1B	20.00 Hz, PW 0.10 ms, Delay 25 μs	pair exact	25 μs , no dispersion; equal widths
2	20.00 Hz, PW 10.00 ms, Delay 10 μs	rising edge exact	10 μs , no dispersion; equal widths
2B	20.00 Hz, PW 0.10 ms, Delay 10 μs	pair exact	10 μs , no dispersion; equal widths
3	20.00 Hz, PW 0.50 ms, Delay 100 μs	pair exact	100 μs , no dispersion
4	20.00 Hz, PW 2.00 ms, Delay 25 μs	rising edge exact	25 μs ; A–B width difference below instrument resolution
5	20.00 Hz, PW 2.00 ms, Delay 1,000 μs	scheduled	dispersion of $\pm 4 \mu s$, independent on the falling edge of A and the rising edge of B
6	20.00 Hz, PW 10.00 ms, Delay 5,000 μs	scheduled	pair fits within the period

In the exact regimes the port B edges are counted in clock cycles from the actual falling edge of port A, so the $\pm 4 \mu s$ quantization of the port A edge is transferred to port B and the delay between them does not vary. In the scheduled regime each edge is placed independently on the 4 μs grid, and the delay carries the corresponding dispersion. All tests in this section: 6,000 pulses fired, 0 timing faults.

2. Cycle constraint

#	Configuration	Menu limit	Measured
7	100.00 Hz (T = 10,000 μ s), PW 4.93 ms requested with a longer Delay	Delay limited to 40 μ s	A–B delay 40 μ s; B to next A 100 μ s
8	10.00 Hz (T = 100,000 μ s), PW 5.00 ms	Delay limited to 89,900 μ s	A–B delay 89.9 ms; B to next A 100 μ s

Test 7: $4,930 + 40 + 4,930 = 9,900 \mu\text{s}$, leaving the 100 μs minimum low time of the cycle.

Test 8: $5,000 + 89,900 + 5,000 = 99,900 \mu\text{s}$, leaving the same 100 μs . The limit displayed by the menu equals $T - 100 \mu\text{s} - 2 \times \text{PW}$ in both cases.

The parameter set first constrains the other: with the pulse width already defined, the Delay editor is limited to the remaining room, and vice versa. The output therefore always matches the displayed values.

Tests 7 and 8: 30,000 and 3,000 pulses fired, 0 timing faults.

3. Port B with the Burst and NPS engines

#	Configuration	Expected	Measured	Faults
9a	Burst 10.00 Hz, Count 10, Gap 1,000 ms, PW 5.00 ms, Delay 25 μ s	1,580 pulses	1,580	0
9b	NPS Count 4, ITlmin 20 ms, PW 5.00 ms, Delay 25 μ s	1,200 pulses, no interval below 20 ms	1,200, constraint met	0

Test 9a: cycle = $(10 - 1) \times 100 \text{ ms} + 5 \text{ ms} + 1,000 \text{ ms} = 1,905 \text{ ms}$; 157 complete cycles in 300 s (1,570 pulses) plus a further complete train of 10 pulses within the remaining 915 ms.

The pulse pair follows every pulse of both engines.

4. Time base integrity

#	Configuration	Expected	Measured	Faults
10	100.00 Hz, PW 0.20 ms, Delay 500 μ s	period 10,000 μ s, no anomalies	no period jumps observed	0

This configuration maximises the interrupt-masked window used to place the pulse pair. The firmware limits that window to 900 μs , below the 1.024 ms overflow period of the system time base, so no overflow can be missed and the period does not drift.

5. Summary

All configurations met their expected result. Pulse counts matched the computed values exactly in Continuous, Burst and NPS modes, and no timing faults were reported in any session, including sessions of 30,000 pulses with port B enabled. The programmed delay was reproduced without measurable dispersion whenever the pulse pair fits in a single interrupt-masked window, and within $\pm 4 \mu\text{s}$ otherwise. The menu enforced the cycle constraint in every case tested, with the oscilloscope confirming the residual 100 μs minimum low time.

Not covered by this report: absolute clock accuracy (the Arduino Uno R3 ceramic resonator is typically within $\pm 0.5 \%$, and all timings scale with it), which requires an external time reference; and the systematic width difference between ports A and B in the intermediate regime, which is below the resolution of the instrument used (up to 4 μs , that is 0.2 % of a 2 ms pulse).